

Twenty-five structural conjectures in number theory

Canonical mechanisms, phase laws, and targeted diagnostics

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Abstract

This manuscript gives exactly twenty-five candidate conjectures in five programmes: connected prime-pattern fields, arithmetic occupancy and class groups, finite logarithms and algebraic tori, arboreal arithmetic dynamics, and marked factorization processes. Each conjecture names a probability space or deterministic averaging operation, identifies a structural mechanism, states a failure mode, and records what a reproducible finite computation does and does not test. The collection is literature-screened but does not claim professionally certified priority. Its aim is to offer conjectures whose resolution would expose new invariants, new local-to-global bridges, or new phase boundaries rather than merely add further special cases of standard heuristics.

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1 Admission standard

A conjecture is admitted here only when it satisfies five requirements.

- A1. Canonical formulation.** The objects, averaging law, normalization and limit are fixed before the data are inspected.
- A2. Mechanism.** The statement identifies an arithmetic or geometric source, not merely a Gaussian, Poisson, Gumbel, Bateman–Horn or Poisson–Dirichlet conclusion.
- A3. Discrimination.** At least two plausible models give different answers.

A4. Leverage. Truth would organize a family of problems, and falsity would identify a missing invariant rather than merely alter a fitted constant.

A5. Proof interface. The statement exposes a first serious intermediate theorem, such as a connected correlation estimate, a large-sieve inequality, a uniform Chebotarev range or a local partition-function calculation.

Several structural commitments hold throughout. Local singular-series and zero-correlation descriptions are conjectured to *match*, not added as independent covariance sources. Motif clusters are connected polymers of arbitrary size, not connected components of a finite difference graph. Exceptional zeros are studied along hypothetical exceptional conductors, since Page’s theorem rules out a positive-density exceptional ensemble. Class-group conditioning retains a genuine family and does not fix the prime-discriminant factors. Toric finite logarithms use prime-to- p torsion lifts, which makes functoriality possible. The arboreal resolution barrier uses total sampling complexity and missing mass rather than any single coordinate discrepancy. Dynamical squarefreeness uses level-dependent local densities rather than a fixed-prime density that must eventually vanish. Multivariate smoothness uses exact p -adic valuation states rather than overlapping root-count sets.

Evidence convention. The companion script `research_quality_25_experiments.py` is deterministic and designed to finish on ordinary hardware, and the measured run time is recorded in its JSON output. Its output is a finite diagnostic, not evidence of proof. A computation that tests only a definition, a null case or a fixed-frequency shadow is labelled accordingly. Numerical ranges are small relative to every asymptotic below.

2 Summary of the twenty-five conjectures

No.	Name	Distinctive content
1	Connected motif cumulant expansion	All fixed-order cumulants are diagram sums over exact overlap types with connected singular-series coefficients.
2	Overlap-rank spectral filtration	The overlap forms define the complete asymptotic eigenspace filtration of finite motif families.
3	Mesoscopic local-spectral reciprocity	Local singular-series and multiple-zero calculations give the same covariance quadratic forms after common smoothing.
4	Prime-motif polymer law	The Palm cluster law is determined by every admissible connected union of translated motifs.
5	Cycle-rank suppression in prime cumulants	Each independent diagram cycle costs a logarithmic order, making hypertrees the first non-Gaussian carriers.
6	Connected occupancy expansion	Least-prime survival has a full expansion whose coefficients are connected same-class prime correlations.
7	Dual autocorrelation laws and Gauss transfer	Additive local correlations and multiplicative zero energies are distinct observables linked by an exact finite transform.

No.	Name	Distinctive content
8	Terminal environment–polymer process	The complete last-class process is conditionally Poisson-clustered in a random low-mode environment.
9	Exceptional-zero hazard transfer	Along a hypothetical exceptional sequence, the first-arrival contrast is the explicit integrated exceptional hazard and has no leading orthogonal harmonics.
10	Entropy–conductor local independence of odd class groups	Cohen–Lenstra statistics persist while both cell entropy and the modulus of the imposed local data remain subcritical.
11	Kummer–Haar law	Fermat-quotient vectors are Haar on the relation torus determined by saturated multiplicative rank.
12	Power-range rank-dimensional large sieve	A positive power range of finite-logarithm frequencies obeys a large-sieve inequality with dual dimension equal to rank.
13	Shrinking-target phase transition for finite logarithms	Saturated rank controls the divergence, critical Poisson regime, and summability of shrinking targets, including Wieferich events.
14	Height-family rank–depth rare-event process	The critical logarithmic process is Poisson, while summable strata retain their exact small-prime Bernoulli atoms.
15	Functorial toric finite logarithm	Prime-to- p torsion lifts produce a functorial first logarithm on every unramified torus.
16	Finite-index universality of arboreal resolution depth	Open arboreal subgroups with comparable ramification have critical observable depths differing by only a bounded amount.
17	Critical arboreal occupancy profile	After analytic error is removed, the complete boundary law is the nonuniform multinomial occupancy process of Haar conjugacy weights.
18	Coloured dynatomic distribution-and-factor process	Uniform local distribution of slowly growing dynatomic components drives independent coloured, Galois-marked factor processes.
19	Uniform squarefree sieve for growing dynatomic degree	Squarefreeness is given by a level-dependent Euler product uniformly while the dynatomic degree grows slowly.
20	Divisor-sensitive dynamical gcd classification	Positive normalized gcd height is exactly the obstruction created by a periodic subvariety carrying a common pulled-back divisor.
21	Frobenius-marked Poisson–Dirichlet process	Macroscopic factors of an irreducible polynomial value carry fixed-point-size-biased Frobenius marks.
22	Residual-mass small–large independence	Small valuations and large marked factors decouple only after the consumed logarithmic mass is removed.
23	Projective p -adic Gibbs gluing for reducible values	Conditional factor processes form a projectively consistent Gibbs mixture generated by exact valuation states.

No.	Name	Distinctive content
24	Adelic saddle-point law for joint smoothness	Archimedean size profiles and exact p -adic valuation integrals jointly determine the moving entanglement factor.
25	Buchstab–Bateman–Horn factor-count flow	One universal family of Buchstab components predicts rough prime, semiprime, and higher-almost-prime polynomial values.

3 Programme I: connected prime-pattern fields

Let $H \subset \mathbb{Z}$ be a finite admissible motif and put

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n+h), \quad Y_H(x, L) = \sum_n w\left(\frac{n-x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)),$$

where w is a fixed smooth compactly supported function. The window position x is sampled from $[X, 2X]$ with density $dx/(x \log 2)$, and $X^\varepsilon \leq L \leq X^{1-\varepsilon}$. For a finite collection of translated motifs, the *exact overlap type* records every equality between their shifted prime constraints. The connected singular series is obtained from the usual singular series by Möbius inversion over set partitions.

Conjecture 1 (Connected motif cumulant expansion). *Fix $r \geq 2$ and admissible motifs $H_1, \dots, H_r$. The joint cumulant $\kappa_r(Y_{H_1}, \dots, Y_{H_r})$ has a uniform asymptotic expansion indexed by exact overlap types τ of r translated motifs. The term attached to τ is the sum, over all translation vectors realizing τ , of*

$$W_\tau(X, L; \mathbf{t}) \mathfrak{S}^c \left(\bigcup_{j=1}^r (H_j + t_j) \right) \prod_v (\log X)^{m_v-1},$$

where m_v is the multiplicity of the prime constraint at the union point v and W_τ is the exact intersection weight of the windows. For every fixed truncation of the ordering by powers of L and $\log X$, the remainder is smaller than the last retained scale, uniformly in the stated mesoscopic range.

Insight. This turns the Hardy–Littlewood singular series into a connected generating functional for a multitype arithmetic field. The overlap poset records equal prime constraints, while partition Möbius inversion records genuine many-body interaction. Neither structure alone is sufficient. The first decisive theorem would establish the formula for third cumulants of two distinct pair motifs. Montgomery–Soundararajan moment methods and Kuperberg’s odd-moment analysis are the closest antecedents [1, 4], with the singular-series sums of Kuperberg, Rodgers and Roditty-Gershon in number fields as the multitype precedent [5].

Finite diagnostic. For four pair fields below $3 \cdot 10^6$, standardized fourth connected cumulants remain of moderate size as the window mean grows. This checks neither the coefficients nor the uniform asymptotic. It only shows that the proposed cumulant object is numerically stable enough to study.

Conjecture 2 (Overlap-rank spectral filtration). *Let $H_1, \dots, H_m$ be fixed motifs and $Y_c = \sum_i c_i Y_{H_i}$. For every shared-point rank $j \geq 1$, let A_j be the limiting overlap form obtained by summing the*

connected local weights over translations for which two motifs share exactly j prime constraints. Put

$$\mathcal{W}_j = \bigcap_{k>j} \ker A_k, \quad \mathcal{W}_{-1} = \{0\}.$$

Then the covariance operator has a complete asymptotic spectral filtration: on every nonzero quotient $\mathcal{W}_j/\mathcal{W}_{j-1}$,

$$\frac{\text{Cov}(Y)}{L(\log X)^j}$$

converges to a nondegenerate quadratic form induced by A_j . On the terminal space $\bigcap_{j \geq 1} \ker A_j$, the covariance divided by its renormalized disjoint-kernel scale converges to a nondegenerate limit. Thus no asymptotic eigenmode can have a variance scale strictly between two adjacent nonzero levels of this filtration.

Insight and proof interface. This is stronger than reading off the largest overlap in one variance formula. It says that the overlap poset supplies the complete asymptotic eigenspace decomposition for any fixed finite motif family. Balanced statistics can therefore be designed to remove successive coincidence ranks and expose genuinely disjoint arithmetic correlation. The first decisive test is a three-motif family for which one overlap form has a rational kernel and the next form is nondegenerate on that kernel. The companion computation finds two overlap ranks for four pair motifs but, correctly, no artificial kernel in that example.

Conjecture 3 (Mesoscopic local–spectral reciprocity). *Remove the exact coincidence terms of Conjecture 2. For a Schwartz function ϕ with $\phi(0) = 0$, define the local quadratic form*

$$\mathcal{Q}_{ij}^{\text{loc}}(\phi; L) = \sum_{t \in \mathbb{Z}} \left[L \int_{\mathbb{R}} w(u) w(u + t/L) du \right] \phi(t/L) \mathfrak{S}^c(H_i, H_j + t),$$

with the standard logarithmic renormalization of the translation sum. Independently, insert the full explicit formulas for every factor $\Lambda(n+h)$ in the smoothed motif covariance and let $\mathcal{Q}_{ij}^{\text{spec}}(\phi; X, L)$ be the resulting quadratic form in the multiple correlations of zeta and Dirichlet- L zeros, using the Mellin transforms of w and ϕ .

For every fixed pair of motifs and every bounded set of Schwartz test functions,

$$\mathcal{Q}_{ij}^{\text{spec}}(\phi; X, L) - \mathcal{Q}_{ij}^{\text{loc}}(\phi; L) = o(\mathcal{V}_{ij}(X, L))$$

uniformly for $X^\varepsilon \leq L \leq X^{1-\varepsilon}$, where $\mathcal{V}_{ij}$ is the first nonzero disjoint covariance scale. The assertion is an equality of the covariance quadratic forms after common smoothing, not a literal identification of the additive Fourier variable with zero ordinates.

Insight, boundary, and failure mode. Goldston–Montgomery-type results relate prime variance to zero pair correlation, while tuple conjectures describe the same covariance through singular series. The proposed reciprocity extends that equivalence to products of shifted von Mangoldt functions and their multiple-zero spectra. It prevents double-counting local and spectral formulas as independent sources. A failure would exhibit a mesoscopic covariance functional visible in neither language. The script checks only finite Wiener–Khinchin consistency and the local input, and cannot test the multiple-zero side. Chan’s refinement of the classical pair-correlation equivalence is the closest one-field antecedent [2].

Conjecture 4 (Prime-motif polymer law). *Fix an admissible motif H . On the integer set of occurrence starts, join two starts when the corresponding translates overlap. A polymer is a finite connected set A of starts for which the complete union $U_A = \bigcup_{t \in A} (H + t)$ is admissible. For $R > 0$, let $\mathcal{F}_{X,L,R}$ be the sigma-field, on the random window-position space, generated by the explicit-formula modes with zero ordinate $|\gamma| \leq RX/L$.*

For every fixed polymer radius, conditionally on $\mathcal{F}_{X,L,R}$ the Palm factorial moment measures of the occurrence process converge to the Hardy–Littlewood densities of all complete unions U_A , and these finite-radius laws are projectively consistent. In the successive limits $X \rightarrow \infty$, then $R \rightarrow \infty$, then polymer radius $\rightarrow \infty$, the Palm cluster is finite almost surely, cluster starts are conditionally Poisson, and

$$\theta(H) = \frac{1}{\mathbb{E}^{\text{Palm}}|A|}.$$

Equivalently, the arithmetic polymer partition function is subcritical for every fixed admissible motif.

Insight. Clusters are not connected components of the finite difference graph $H - H$: for $H = \{0, 6\}$ the starts 0, 6, 12 form a connected three-occurrence polymer with admissible union $\{0, 6, 12, 18\}$, so polymers of every size carry weight. The explicit-formula sigma-field fixes the conditioning canonically. The conjecture predicts the entire Palm cluster law and gives a concrete proof interface through convergence of polymer partition functions.

Conjecture 5 (Cycle-rank suppression in prime cumulants). *In the exact connected expansion of Conjecture 1, contract every forced coincidence and associate to each remaining translation diagram its cyclomatic number g , the number of independent cycles in its overlap hypergraph. After all larger coincidence scales are removed, a diagram of cyclomatic number g is smaller than a hypertree diagram with the same vertices by at least $(\log L)^{-g}$. Consequently the first nonzero odd cumulant of any balanced motif statistic is carried by connected acyclic overlap hypergraphs. For the single-prime motif this recovers Kuperberg’s predicted order*

$$\kappa_{2m+1} \asymp L^m (\log L)^{m+1},$$

with leading constant equal to the total connected hypertree weight.

Insight. A Gaussian limit only says that normalized odd cumulants vanish. This conjecture identifies the topological invariant governing the first non-Gaussian term: every independent diagram cycle costs one logarithmic order. It therefore predicts which subfamilies of singular-series sums must be evaluated and supplies a mechanism for the odd-moment scale rather than repeating it. Kuperberg’s conjecture and bounds are the single-field calibration [4]. The motif-dependent cycle filtration is the new content.

4 Programme II: arithmetic occupancy and local–global decoupling

For a prime modulus q and $a \in \mathbb{F}_q^\times$, let $p(q, a)$ be the least prime in the class a and put

$$T_q(a) = \frac{\text{Li}(p(q, a))}{q - 1}.$$

The probability space chooses q uniformly among the primes in $[Q, 2Q]$ and then chooses a uniformly from $\mathbb{F}_q^\times$.

Conjecture 6 (Connected occupancy expansion). *For every fixed $R \geq 2$ and t in a fixed compact subset of $[0, \infty)$,*

$$-\log \mathbb{P}[T_q(a) > t] = t + \sum_{r=2}^R \frac{A_r(t)}{(\log Q)^{r-1}} + o((\log Q)^{1-R}).$$

The coefficient A_r is the first-arrival transform of the connected r -prime correlation for primes lying in one residue class, averaged over the dyadic modulus block. No assertion that A_2 alone gives the first correction is built into the definition.

Insight. This is a cluster expansion for an arithmetic coupon collector. It distinguishes the prime process from an exchangeable collector and makes higher connected correlations separately measurable. Li–Pratt–Shakan introduced the occupancy viewpoint for the extreme least prime [6]. The new content is the full survival functional. In the finite computation, the scaled mean speedup rises from about 1.27 to 1.60 across the sampled modulus bands, while the cover time remains roughly 2 to 2.5 below the iid harmonic benchmark. The range cannot distinguish $1/\log Q$ from nearby slowly varying scales.

Conjecture 7 (Dual autocorrelation laws and the Gauss transfer). *Extend the centered least-prime field to $f_q : \mathbb{F}_q \rightarrow \mathbb{R}$ by assigning a fixed recorded value at 0 and then subtracting its full mean. Define normalized additive and multiplicative coefficients*

$$A_q(\xi) = q^{-1/2} \sum_{a \bmod q} f_q(a) e(-a\xi/q), \quad M_q(\chi) = (q-1)^{-1/2} \sum_{a \neq 0} f_q(a) \overline{\chi(a)}.$$

For every nontrivial character, the exact Gauss identity gives

$$M_q(\chi) = \frac{\tau(\overline{\chi})}{\sqrt{q(q-1)}} \sum_{\xi \neq 0} A_q(\xi) \chi(\xi).$$

The conjectural content is the following pair of compatible laws.

(i) *For every fixed additive lag h , the dyadically averaged autocorrelation $q^{-1} \sum_a f_q(a) f_q(a+h)$ has a full expansion whose coefficients are the first-arrival transforms of the connected same-class prime correlations of Conjecture 6.*

(ii) *For every fixed character order and compatible family of characters, the dyadically averaged energies $|M_q(\chi)|^2$ and their finite joint moments are given by the low-zero explicit-formula correlations of the corresponding Dirichlet L -functions.*

These two laws obey the exact Gauss-transfer identity above at every q . No claim that additive and multiplicative mechanisms are orthogonal or exhaustive is made.

Insight and diagnostic. Additive and multiplicative Fourier systems are not independent coordinate worlds: the Gauss transform converts one coefficient vector into the other exactly. The number-theoretic claim is instead that two different autocorrelation observables isolate two mechanisms while remaining algebraically compatible. A projection of the tensor square onto additive-difference and multiplicative-ratio matrix algebras cannot serve here: in direct computation such subspaces explain only a small fraction of the rank-one energy. The script verifies the Gauss identity numerically and records the two autocorrelation families separately.

Conjecture 8 (Terminal environment–polymer process for the last classes). *Let b_Q be the unique centering satisfying*

$$\mathbb{E}_{q \in [Q, 2Q]} \# \{a : T_q(a) > b_Q\} = 1,$$

and put $\Xi_q = \sum_{a \neq 0} \delta_{T_q(a) - b_Q}$. For $\eta > 0$, let $\mathcal{G}_{Q,\eta}$ be the sigma-field generated, under the random-modulus law, by those normalized multiplicative-character coordinates of the arrival field whose dyadic variance is at least η . Define the conditional log-intensity field

$$W_{Q,\eta}(x) = \log \mathbb{E}[\#\{a : T_q(a) > b_Q + x\} \mid \mathcal{G}_{Q,\eta}] + x.$$

First let $Q \rightarrow \infty$ and then $\eta \downarrow 0$. The fields $W_{Q,\eta}$ converge in this successive limit to a random locally bounded field W .

Conditionally on $W_{Q,\eta}$, the cluster starts of Ξ_q converge to a Poisson process with intensity

$$\theta_{\text{occ}} \exp\{-x + W(x)\} dx.$$

The within-cluster law is the projective connected-polymer law generated by the terminal connected occupancy coefficients of Conjecture 6. Removing the conditioning gives the corresponding Cox polymer process. Both the random environment and the cluster law are part of the limit, and neither is compressed into a fitted location shift.

Insight. The statement predicts the joint law, multiplicities, character profiles, and spacings of all final residue classes. It discriminates among an iid coupon collector, a scalar Cox mixture, and a genuinely field-valued arithmetic environment. The variance cutoff above makes the low-mode field canonical rather than selecting modes after inspecting an extreme. The implicit centering avoids extrapolating a compact-time expansion to $t \asymp \log Q$.

Conjecture 9 (Exceptional-zero hazard transfer). *Suppose that primitive real characters χ_j of conductors $q_j \rightarrow \infty$ have real zeros β_j with $(1 - \beta_j) \log q_j \rightarrow 0$. Let $x_j(t)$ be defined by*

$$\text{Li}(x_j(t))/(q_j - 1) = t,$$

and let $S_j^\pm(t)$ be the survival proportions of residue classes with $\chi_j(a) = \pm 1$. After subtracting the ordinary connected occupancy contribution of Conjecture 6, put

$$H_j(t) = 2 \int_0^t x_j(u)^{\beta_j - 1} du.$$

For every fixed $T > 0$,

$$\sup_{0 \leq t \leq T} \left| \log S_j^+(t) - \log S_j^-(t) - H_j(t) \right| \rightarrow 0.$$

Furthermore every character projection orthogonal to 1 and χ_j is $o(\|H_j\|_{C[0,T]})$. Thus the exceptional explicit-formula term remains rank one under the nonlinear first-arrival map to leading order.

Insight and conditional status. Page's theorem allows at most one exceptional character below a conductor bound, so no positive-density exceptional ensemble is available [7]. The formula above uses a moving, explicitly normalized target rather than saying that a sequence-dependent quantity “converges to itself.” If an exceptional sequence exists, higher harmonics would pinpoint exactly where nonlinear first-arrival dynamics defeats the rank-one explicit-formula picture. If no such sequence exists, the conjecture is intentionally instanceless and is not counted as evidence for exceptional zeros.

Conjecture 10 (Entropy–conductor local independence of odd class groups). *Fix an odd prime ℓ . Sample negative fundamental discriminants $D \in [-2Y, -Y]$ and condition on a fixed coarse two-primary invariant. Let S_Y be a set of primes not dividing D , let $M_Y = \prod_{p \in S_Y} p$, and prescribe*

an attainable vector of splitting signs (D/p) . Write H_Y for the logarithm of the number of attainable sign cells after reciprocity relations are imposed.

Uniformly over cells whose expected size tends to infinity, the conditional law of $\text{Cl}(D)[\ell^\infty]$ converges in total variation to the unconditioned Cohen–Lenstra law whenever, for some fixed $\varepsilon > 0$,

$$H_Y \leq (1 - \varepsilon) \log Y, \quad \log M_Y \leq (\tfrac{1}{2} - \varepsilon) \log Y.$$

More generally the same conclusion holds whenever the local-cell entropy and the analytic conductor complexity jointly lie in the range in which discriminants are equidistributed among the corresponding sign cells. At the information boundary $H_Y \geq (1 + \varepsilon) \log Y$, no uniform total-variation assertion is possible.

Insight and literature boundary. The first constraint prevents empty information cells, and the second records that the same number of sign bits can be imposed at radically different moduli. The conjecture is therefore not merely “independence under many local conditions,” but a two-resource local–global principle. Wood’s fixed-local-condition results and conjectures are the nearest foundation [8]. A proof in any range with $|S_Y| \rightarrow \infty$ would be a substantial advance and would have direct Selmer- and ray-class analogues.

5 Programme III: finite logarithms, rank and algebraic tori

For $p \nmid a$, write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a tuple $g = (g_1, \dots, g_s)$, let $R(g)$ be its multiplicative relation lattice and let $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$ be the identity component of its annihilator. Its dimension is the saturated multiplicative rank.

Conjecture 11 (Kummer–Haar law for Fermat quotients). *Let $\Gamma \leq \mathbb{Q}^\times$ be saturated of rank r , generated by g . As p varies in any fixed compatible cyclotomic–Kummer Chebotarev class,*

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p}$$

equidistributes with Haar measure on $\mathbb{T}_g$. Equivalently, every Fourier coefficient outside the saturated relation lattice tends to zero, and no proper closed subgroup of $\mathbb{T}_g$ contains the support.

Insight. The controlling invariant is saturated rank, not the number of displayed bases. The statement unifies marginal Fermat-quotient distribution, simultaneous near-zero events and restrictions to fixed splitting classes. Existing exponential-sum estimates treat different horizontal or averaged regimes [9]. The full prime-varying relation-torus law is substantially stronger. Below 120000, exact relations among bases 2, 3, 6, 12 hold to machine precision and fixed-frequency Fourier coefficients are small. Only the elementary support and fixed-frequency shadows are tested.

Conjecture 12 (Power-range rank-dimensional large sieve for finite logarithms). *For every saturated $\Gamma \leq \mathbb{Q}^\times$ of rank r , choose a Smith-normal-form basis of the character lattice $X^*(\mathbb{T}_g) = \mathbb{Z}^s / R(g)^{\text{sat}}$ and use the associated sup norm. Changing this basis alters heights by bounded factors only. There*

exists $\delta_\Gamma > 0$ such that, for every $\varepsilon > 0$, every $M \leq X^{\delta_\Gamma}$, and all complex weights a_p supported on $X < p \leq 2X$,

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \leq M}} \left| \sum_{X < p \leq 2X} a_p e\left(\frac{\langle m, q_p(g) \rangle}{p}\right) \right|^2 \ll_{\Gamma, \varepsilon} (M^r + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The estimate remains valid after conditioning on fixed compatible cyclotomic–Kummer data. The exponent δ_Γ is positive and invariant under changing the generating tuple, but no near-full frequency range is assumed.

Insight and calibration. This is the quantitative engine behind the fixed-scale Haar law, with the dual dimension equal to saturated rank. Existing Fermat-quotient exponential sums concern different horizontal or averaged regimes [9]. A positive power range would already control genuinely shrinking targets. The explicit character-lattice basis gives the frequency height a well-defined meaning on the quotient group.

Conjecture 13 (Shrinking-target phase transition for finite logarithms). *Let $\Gamma \leq \mathbb{Q}^\times$ be saturated and torsion-free of rank r . Choose a basis of its relation torus. For $0 < \alpha < 1$, fixed $B > 0$, and a prime p , let $E_p(\alpha, B)$ be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most $\lfloor Bp^{1-\alpha} \rfloor$. Then*

$$\#\{p \leq X : E_p(\alpha, B)\} \sim \sum_{p \leq X} \left(\frac{2\lfloor Bp^{1-\alpha} \rfloor + 1}{p} \right)^r$$

whenever $\alpha r < 1$, whereas only finitely many such primes occur when $\alpha r > 1$.

At the critical line $\alpha r = 1$, choose U_S uniformly from $[0, S]$ and put

$$\Xi_S = \sum_{\substack{E_p(\alpha, B) \text{ occurs} \\ 0 \leq \log \log p \leq S}} \delta_{\log \log p - U_S}.$$

This random translated process converges locally to a Poisson process of intensity $(2B)^r$. Separately, the exact target $q_p|_\Gamma = 0$ has Haar mass p^{-r} : its $\log \log p$ point process has unit critical intensity at rank one and is finite at rank at least two.

Insight and discrimination. The conjecture extends Conjecture 11 from fixed sets to targets shrinking at a power of the residue modulus and makes rank the phase-transition invariant. Exact Wieferich events are the lattice-scale endpoint, not an isolated folklore assertion. The Poisson clause supplies a legitimate probability space by random translation in $\log \log p$ -time. It sharply distinguishes the Haar shrinking-target model from heuristics predicting much rarer fixed-base Wieferich events, such as those discussed by Gras [3] and, through the mod p Leopoldt lens, by Boeckle, Guiraud, Kalyanswamy and Khare [10].

Conjecture 14 (Height-family rank–depth rare-event process). *Choose an ordered r -tuple of integers independently and uniformly from $[-H, H] \setminus \{0\}$, conditioned on multiplicative independence. For a prime p and $k \geq 1$, mark (p, k) when every coordinate is a p -adic unit and satisfies $a^{p-1} \equiv 1 \pmod{p^{k+1}}$. Its exact local probability is*

$$(1 - p^{-1})^r p^{-rk},$$

and, conditional on depth k , the probability of depth $k + 1$ is p^{-r} .

For the critical case $rk = 1$, fix $\varepsilon > 0$ and form

$$\Xi_H = \sum_{p^{k+1} \leq H^{1-\varepsilon}} \mathbf{1}_{(p,k) \text{ marked}} \delta_{\log p / \log H}.$$

On every compact subinterval of $(0, (1 - \varepsilon)/(k + 1))$, Ξ_H converges to a Poisson process of intensity dt/t .

For $rk > 1$, the full marked set converges in finite-prime marginals to the product of the exact local Bernoulli laws. After deleting primes $p \leq z$ and then taking $H \rightarrow \infty$, its large-prime tail is Poisson with mean $\sum_{p > z} (1 - p^{-1})^r p^{-rk}$, and this mean tends to zero as $z \rightarrow \infty$.

Insight. No single undifferentiated Poisson process can describe both regimes: the rank-depth product $rk = 1$ has logarithmically many events, whereas $rk > 1$ is a summable collection with non-negligible small-prime atoms. The statement gives each regime its natural limit and retains the exact Haar-lift dependence between depths. The range $p^{k+1} \leq H^{1-\varepsilon}$ is where the height box genuinely samples the required residue classes.

Conjecture 15 (Functorial finite logarithms on algebraic tori). *Let $T/\mathbb{Q}$ be a torus with good unramified reduction at p , and let $P \in T(\mathbb{Q}_p)$ be integral. The finite group $T(\mathbb{F}_p)$ has order prime to p . Let $\tau_p(\bar{P})$ be the unique prime-to- p torsion lift of the reduction of P . Define*

$$\ell_p(P) = \frac{1}{p} \log_T \left(P \tau_p(\bar{P})^{-1} \right) \pmod{p} \in \text{Lie}(T)(\mathbb{F}_p).$$

If S is the identity component of the Zariski closure of $\langle P \rangle$, then, after choosing an integral basis of $\text{Lie}(S)$ and dividing its residue representatives by p , the resulting points equidistribute with Haar measure on $(\mathbb{R}/\mathbb{Z})^{\dim S}$ as p varies through every compatible Frobenius class. For every morphism of unramified tori $\phi : T \rightarrow T'$ one has

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)).$$

Depth- k vanishing therefore obeys the same codimension phase law as Conjecture 13, with r replaced by $\dim S$.

Insight. Only chosen integer representatives are scaled to the real torus, so no element of $\text{Lie}(T)(\mathbb{F}_p)$ is ever divided by p . Removing the prime-to- p torsion lift is canonical and functorial, unlike multiplication by unrelated source and target exponents. For $\mathbb{G}_m$ this recovers the Fermat quotient up to sign. The script verifies several functorial identities exactly. The horizontal Haar law is the conjectural content.

6 Programme IV: arboreal resolution and dynamical factors

Let $K_n = \mathbb{Q}(f^{-n}(a))$, $G_n = \text{Gal}(K_n/\mathbb{Q})$, and let $\mu_n(C) = |C|/|G_n|$ on the conjugacy classes of G_n . Put $N_X = \pi(X)$ and define the canonical iid sampling functional by

$$\mathcal{S}(n, X) = \mathbb{E} \text{TV} \left(\frac{1}{N_X} \sum_{j=1}^{N_X} \delta_{Z_j}, \mu_n \right), \quad Z_j \stackrel{\text{iid}}{\sim} \mu_n.$$

It is explicitly computable from μ_n . The usual bound $\mathcal{S} \ll \frac{1}{2} \sum_C \min\{\mu_n(C), \sqrt{\mu_n(C)/N_X}\}$ is used only as a proxy in the finite script. If k_n is the number of conjugacy classes, define the GRH-scale aggregate conductor term

$$\mathcal{A}(n, X) = \frac{\sqrt{k_n|G_n|} [\log |D_{K_n}| + [K_n : \mathbb{Q}] \log X] \log X}{\sqrt{X}}.$$

Conjecture 16 (Finite-index universality of arboreal resolution depth). *For an arboreal representation with finite quotient G_n , let $\mathcal{S}_G(n, X)$ be the expected total variation of the empirical measure of $\pi(X)$ iid samples from Haar conjugacy-class measure on G_n . Let $\mathcal{A}_f(n, X)$ be the explicit aggregated GRH-scale Chebotarev functional defined immediately above from k_n , $|G_n|$, the field degree, and $\log |D_{K_n}|$. Define the model resolution depth*

$$n_f^*(X) = \max\{n : \mathcal{S}_G(n, X) + \mathcal{A}_f(n, X) \leq 1/10\}.$$

Consider two non-PCF maps of the same degree whose arboreal images are open and whose preimage-field discriminants have comparable ramification entropy, in the sense that $\log |D_{K_n^{(1)}}| / \log |D_{K_n^{(2)}}|$ stays between two positive constants. Then

$$n_{f_1}^*(X) - n_{f_2}^*(X) = O(1).$$

The actual largest level at which empirical Frobenius measures converge in total variation differs from $n_f^(X)$ by $O(1)$ in the same family-averaged regime.*

Insight. An open subgroup contains a fixed level stabilizer, so finite index changes only bounded initial tree data, while the doubly exponential state growth fixes the leading observable depth. The ramification comparison is essential: group index alone cannot control discriminants. This is a universality statement about how much of an infinite Galois representation is observable, rather than a repackaged effective-Chebotarev implication. Large arboreal images and effective Chebotarev provide the inputs [12, 11]. The script compares iid resolution functionals for the full binary wreath model and an explicit index-two parity subgroup at small levels.

Conjecture 17 (Critical arboreal occupancy profile). *Sample the parameter uniformly from a one-parameter family of height at most B whose arboreal images have uniformly bounded index, and let $n = n(B)$ and $X = X(B)$ tend to infinity. Suppose the family-averaged analytic Chebotarev error is $o(1)$ in total variation. Write $\mu_n(C) = |C|/|G_n|$ and $N_X = \pi(X)$. Then the complete conjugacy-cylinder count vector has the same asymptotic law as the multinomial occupancy vector obtained from N_X independent samples from μ_n .*

Equivalently, after Poissonizing the number of sampled primes, counts in classes with $N_X \mu_n(C) \rightarrow \lambda_C$ converge jointly to independent $\text{Pois}(\lambda_C)$ variables, while the empirical missing mass, collision mass, and total variation converge to the corresponding nonuniform occupancy functionals. De-Poissonization recovers the fixed-total multinomial law. In particular the total-variation transition occurs exactly when the canonical iid sampling functional ceases to vanish.

Insight and risk. The conjecture predicts the whole critical profile rather than only a threshold. The actual Haar class weights, not $|G_n|$ treated as equiprobable states, determine the boundary. Its falsity would reveal prime-to-prime or parameter-to-parameter correlations surviving after every effective Chebotarev error has been removed. The companion calculation evaluates the exact iid missing-mass and total-variation proxies at fixed small levels. Growing conjugacy depth remains inaccessible.

Conjecture 18 (Coloured dynatonic distribution-and-factor process). *For $f_c(x) = x^d + c$, let the critical dynatonic polynomial at level n factor as $\Psi_n = \prod_{j=1}^{s_n} F_{n,j}$ over $\mathbb{Z}$, and sample $|c| \leq H$ away from the PCF, reducible, discriminant, and fixed-resultant loci.*

For every fixed n , after conditioning on each fixed finite local state and normalizing each component by its residual logarithmic mass, the component-coloured prime-factor processes converge jointly to independent PD(1) processes. Unramified factors in colour j carry the fixed-point-size-biased Frobenius mark in the splitting field of $F_{n,j}$, and the total logarithmic mass of ramified factors is $o(1)$ in probability.

Uniformly for $n = n(H)$ with $\deg \Psi_n = o(\log H)$, the same conclusion holds at the level of every fixed correlation functional whose support lies inside the level of distribution supplied by the exact local root-count tensors of the $F_{n,j}$. Thus the fixed-level statement is a full process law, while the growing-level statement makes only the restricted-support assertion justified by its distribution range.

Insight. The statement separates two mathematically different claims: a small positive distribution level does not automatically give a full Poisson–Dirichlet process for growing degree. Dynatonic components are genuine colours arising from exact-period geometry, and their Galois marks distinguish families with similar unmarked factor sizes. Bharadwaj–Rodgers supply the factor-process template and the distinction between full and restricted-support conclusions [15]. The coloured dynatonic and growing-degree uniformity are the new claims.

Conjecture 19 (Uniform squarefree sieve for growing dynatonic degree). *For each n and prime p , let*

$$\delta_{n,p} = p^{-2} \# \{c \bmod p^2 : p^2 \mid \Psi_n(c) \text{ and } p \nmid v_m(c) \text{ for } m < n\}.$$

Here $v_m(c) = f_c^m(0)$ denotes the m th critical-orbit value, so the condition selects primitive divisibility in the sense of the primitive-prime-divisor literature [13]. In the regime $\deg \Psi_n = o(\log H)$,

$$\mathbb{P}_{|c| \leq H}[\Psi_n(c) \text{ is squarefree}] = \prod_p (1 - \delta_{n,p}) + o(1)$$

uniformly in n , after explicitly deleting the finitely many critical-collision local states. Moreover the logarithm of the squarefull part is $o(\log |\Psi_n(c)|)$ in probability.

Insight. For fixed p , a first primitive hit occurs by bounded time modulo p^2 , so no nonzero fixed-prime density can stabilize as $n \rightarrow \infty$: the local density must depend on n . For levels 2, 3, 4, empirical squarefree fractions 0.622, 0.916, 0.987 compare with truncated local products 0.611, 0.924, 0.989, and the agreement checks the level-dependent local object at small degree only.

Conjecture 20 (Divisor-sensitive classification of dynamical gcd height). *Let f, g be disintegrated polynomials of the same degree, let $F = (f, g)$ on $(\mathbb{P}^1)^2$, and let $P = (a, b)$ have positive canonical heights. Let E be the exceptional divisor of the blow-up at $(0, 0)$, so that the generalized logarithmic gcd is the E -height of $F^n(P)$.*

The normalized gcd height has positive limsup if and only if, after replacing F by an iterate, P lies on an F -periodic irreducible curve Z such that on the normalization of Z the pullbacks of the two coordinate-zero divisors share a nonzero effective component. In the absence of such a divisor-compatible periodic curve,

$$\frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} \longrightarrow 0.$$

The converse is asserted in the same range in which the Vojta inequality for the blow-up controls proximity to E . Without that Diophantine input it is a separately identified conditional clause.

Insight and current boundary. A periodic correspondence alone is insufficient: it must carry the actual divisors measured by gcd height. This formulation joins dynamical unlikely intersections, blow-up heights, and invariant-curve classification in one obstruction. Recent work of Yang–Zhong resolves several set-theoretic common-zero and dynamical-GCD questions [14]. The height-level divisor-compatible if-and-only-if is the proposed additional boundary, not a restatement of those results.

7 Programme V: Frobenius-marked factorization and sieve flow

Conjecture 21 (Frobenius-marked Poisson–Dirichlet factor process). *Let $f \in \mathbb{Z}[x]$ be irreducible with splitting field L and transitive Galois group G . Sample n uniformly from a dyadic interval $[N, 2N]$. Mark each unramified prime factor p of $f(n)$ by its conjugacy class $C \subset G$ and give it mass $u = \log p / \log |f(n)|$. The macroscopic process converges to a marked PD(1) process with mark law*

$$\mathbb{P}(C) = \frac{|C|}{|G|} \text{fix}(C).$$

Conditional on the masses and on any fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with its induced permutation representation, the marked process pushes forward functorially.

Insight. Chebotarev supplies $|C|/|G|$, divisibility by a polynomial value size-biases by the number of fixed roots, and Burnside’s lemma normalizes the law. This joins Galois theory to the large-factor process. Bharadwaj–Rodgers prove Poisson–Dirichlet limits or restricted-support correlations for well-distributed sequences, including irreducible polynomial values under the relevant distribution hypotheses [15]. For $x^3 - 2$ and $n \leq 1600$, the largest-factor marks split 49.75% versus 50.25% between the one- and three-fixed-point classes, close to the size-biased S_3 prediction.

Conjecture 22 (Small–large independence after residual-mass normalization). *Let $y \rightarrow \infty$ with $\log y = o(\log N)$. Write $S_y(f(n))$ for the y -smooth part and $R_y = f(n)/S_y(f(n))$. The sigma-field generated by $\{v_p(f(n)) : p \leq y\}$ is asymptotically independent of the Frobenius-marked large-factor process of R_y when every large-factor mass is normalized by $\log |R_y|$. Equivalently, the joint Laplace functional factors after this residual-mass normalization and after conditioning on the finite ramification state.*

Insight. Raw independence is false because small factors consume mass. In the finite $x^3 - 2$ sample, the correlation between the number of factors at most 100 and the raw largest share is -0.434 , falling to about 0.107 after residual normalization. The remaining correlation is large enough that the asymptotic independence is not being assumed from the data.

Conjecture 23 (Projective p -adic Gibbs gluing for reducible polynomial values). *Let $f = \prod_{i=1}^s g_i$ be squarefree with irreducible components and sample $n \in [N, 2N]$. For each prime p let ν_p be the exact probability law of the valuation vector $(v_p(g_1(x)), \dots, v_p(g_s(x)))$ for Haar-random $x \in \mathbb{Z}_p$, with partition function*

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} dx.$$

For $y \rightarrow \infty$ with $\log y = o(\log N)$, condition on the complete local state $\mathcal{L}_y = (v_p(g_i(n)))_{p \leq y, i \leq s}$ and normalize each component’s large factors by its own residual logarithmic mass. The coloured marked factor processes are then asymptotically independent.

Before conditioning, their joint law is the projective Gibbs mixture obtained by integrating those independent conditional laws against $\bigotimes_{p \leq y} \nu_p$ and then taking $N \rightarrow \infty$ followed by $y \rightarrow \infty$. Equivalently every bounded finite-dimensional Laplace functional is the limit of the corresponding finite- y mixture, and this projective family is completely determined by the Z_p . No additional colour coupling survives in the limit.

Insight. A scalar Euler factor can correct one avoidance probability but cannot specify a point process. The projective Gibbs formulation encodes every local valuation and the random residual masses they consume. It supplies a functorial rule under multiplying or refactoring polynomials. The strong final sentence is directly falsifiable by a persistent archimedean or medium-prime colour correlation after the iterated limit.

Conjecture 24 (Adelic saddle-point law for simultaneous smooth polynomial values). *Let $f_1, \dots, f_s$ be pairwise coprime primitive polynomials and sample $n \in [N, 2N]$. Put $t = n/N$. For each prime define the exact local partition function*

$$Z_{p,\mathbf{y}}(\boldsymbol{\sigma}) = \int_{\mathbb{Z}_p} p^{-\sum_{i: p \leq y_i} \sigma_i v_p(f_i(x))} dx.$$

For each $t \in [1, 2]$, let $\boldsymbol{\sigma}(t)$ be the unique saddle in the admissible region satisfying

$$\frac{\partial}{\partial \sigma_i} \left(\sum_{p \leq y_i} \log Z_{p,\mathbf{y}}(\boldsymbol{\sigma}) + \sum_j \sigma_j \log |f_j(Nt)| \right) = 0 \quad (1 \leq i \leq s).$$

Uniformly when the smoothness parameters $u_i(t) = \log |f_i(Nt)| / \log y_i$ remain in a fixed compact saddle-point range, the joint smooth-value probability is asymptotic to the integral over $t \in [1, 2]$ of the product of the marginal saddle densities multiplied by

$$\mathcal{E}(\boldsymbol{\sigma}(t)) = \prod_p \frac{Z_{p,\mathbf{y}}(\boldsymbol{\sigma}(t))}{\prod_i Z_{p,i,y_i}(\sigma_i(t))}.$$

Here Z_{p,i,y_i} is the one-polynomial partition function obtained from $Z_{p,\mathbf{y}}$ by retaining the factor f_i alone. After both this archimedean integration and the moving p -adic Euler factor are included, the remaining relative error tends to zero.

Insight. The conjecture is adelic rather than purely Eulerian. It includes exact higher valuations, the variation of polynomial sizes across a dyadic interval, and the movement of the local law under the smoothness tilt, so no assertion that all global correlation is p -adic is smuggled in. Single-polynomial smooth-value estimates are already difficult [16, 17], and a multivariate adelic saddle law would be a major new local–global bridge.

Conjecture 25 (Buchstab–Bateman–Horn factor-count flow). *Let $f \in \mathbb{Z}[x]$ be primitive, irreducible, and admissible, and sample $n \in [N, 2N]$. For y with $u_n = \log |f(n)| / \log y$ in a fixed compact subset of $(1, \infty)$, let $\Omega_y(f(n))$ be the number of prime factors of $f(n)$ exceeding y , counted with multiplicity, on the event $P^-(f(n)) > y$.*

Define universal Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \quad \omega_j(u) = \frac{1}{u(j-1)!} \int_{t_1 + \dots + t_{j-1} \leq u-1}^{t_1, \dots, t_{j-1} \geq 1} \frac{dt_1 \cdots dt_{j-1}}{t_1 \cdots t_{j-1}} \quad (j \geq 2),$$

with an empty integral interpreted as zero. These satisfy $\sum_{j \geq 1} \omega_j(u) = \omega(u)$. Then for every fixed j ,

$$\mathbb{P}(P^-(f(n)) > y, \Omega_y(f(n)) = j) = V_f(y) e^\gamma \mathbb{E}_{n \in [N, 2N]} \omega_j(u_n) + o(V_f(y)),$$

uniformly on compact u -ranges. Summing over j gives the rough-value Buchstab law. The $j = 1$ component and $V_f(y) \sim C_f e^{-\gamma} / \log y$ give the Bateman–Horn prime-value density, while $j = 2$ gives the semiprime transition. If $\max_{N \leq n \leq 2N} |f(n)| < y^2$, every rough value belongs to the $j = 1$ component exactly.

Insight and diagnostic. Primes, semiprimes, and all higher almost-prime transitions are now parts of one process, not a conditional probability obtained by dividing two assumed asymptotics. The polynomial local factors enter only through V_f , while the remaining-factor count is a universal finite- u flow. This is strongly falsifiable: even if total roughness follows Buchstab, the distribution among j can fail. The script compares the observed component vector for $n^2 + 1$ with both the displayed universal profile and a matched random-integer sample, and checks the pointwise endpoint.

8 Quality audit and research priorities

The collection is not flat. The strongest parent mechanisms are Conjectures 1, 3, 4, 6, 10 to 13, 15 to 17, 21 and 23 to 25. The remaining conjectures are structural consequences or extensions that introduce a distinct invariant: overlap rank, cluster polymers, hypertree corrections, terminal occupancy, exceptional spectral response, rank-depth processes, dynatomic colouring, growing-degree squarefreeness, divisor-sensitive gcd geometry and residual-mass decoupling.

The first decisive targets are deliberately smaller than the full statements:

1. Prove the third-cumulant diagram formula for two pair motifs and identify the connected disjoint kernel.
2. Derive $A_2(t)$ in Conjecture 6 and show whether $A_3(t)$ is genuinely one logarithmic order smaller.
3. Prove a rank-one version of the large-sieve inequality Conjecture 12 for $M = X^\delta$.
4. Prove bounded-level stability of the arboreal sampling functional under one explicit open subgroup and compare the resulting critical depths.
5. Prove a two-colour restricted-support version of Conjecture 21 or Conjecture 23.
6. Prove the $j = 1$ and $j = 2$ component laws of Conjecture 25 in an averaged polynomial family before attempting the full fixed-polynomial flow.

A conjecture should be removed rather than defended if its canonical object fails. The collection accordingly contains no claim that a finite computation certifies novelty, no probability on a deterministic sequence without an ensemble, no positive-density Siegel-zero family, no functorial toric logarithm based on incompatible group exponents, and no scalar Euler factor pretending to specify a full point-process interaction.

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